Artificial Intelligence Bias Regulation Aggregator

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Codebase

Background

- The **US government** has taken a **distributed/fragmented** approach to **AI regulation** thus far.
- Some states have enacted their own laws and regulations, while other federal agencies are actively working toward national frameworks.
- **AI-based tools** are rapidly proliferating across the United States, for example, in **high-stakes domains** such as predictive policing, child maltreatment risk assessments, and emotion recognition.
- These tools have been met with widespread **contention**, with researchers and community members alike raising concerns around **bias and fairness**.
- The goal of this project is to understand **how federal and state-level agencies currently regulate their uses of AI, to prevent and remedy *biased* and *discriminatory* patterns.**

Research Questions

Focusing on publicly available information on gov websites:

- What fraction of them contain **relevant** information?
- Among those, how many contained **concrete processes and guidelines** to guard against AI/ML bias?

Results & Discussions

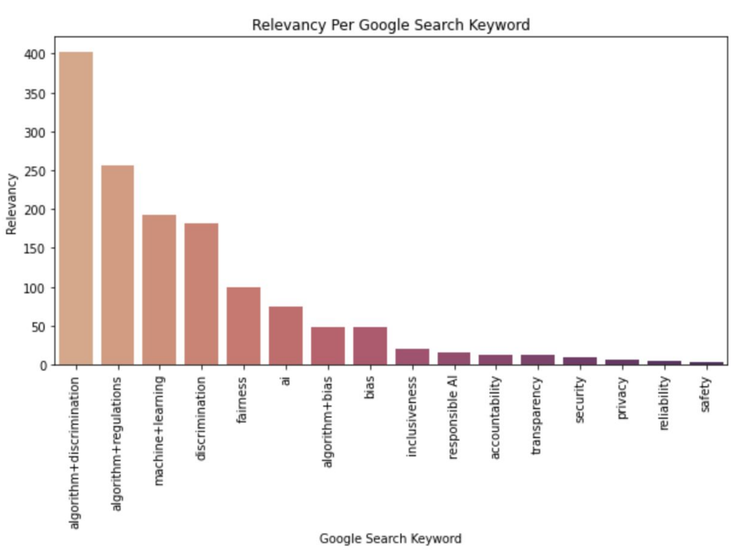

Figure 1. Queries vs. Frequency of Keywords Mentioned
Determined the best Google Search Keywords based on the frequency of the Keywords Mentioned from the URLs fetched.

Methods

Definitions

1. **Search keywords**: bias, discrimination, fairness, algorithm bias, algorithm discrimination, algorithm regulations, accountability, transparency, reliability, safety, privacy, security, inclusiveness, responsible AI, Machine Learning, AI
2. **Keywords mentioned**: bias, discrimination, fairness, algorithm, algorithmic bias, AI, Artificial Intelligence, algorithmic discrimination, algorithm regulations, Machine Learning, ML, inclusive, diverse, diversity

Data Collection:

1. Identify keywords to get the initial set of websites to scrape
2. Web crawling & scraping starting with the seed set, advanced search for .gov sites

Data Analysis:

1. **Quantitative Analysis: Scoring pages** according to their relevance
   a. For each of the pages, we scored based on the frequency of containing the *keywords mentioned.*
2. **Qualitative analysis: Analyze** the top 10 manually.
   a. **Descriptive statistics** with counts and visualizations of most common relevant keywords and portion of keywords that relate to AI bias or other topics.
   b. **Qualitative analysis** of the scraped top K=10 documents

Future Experimentation

- Expanding the scope to other countries, e.g., .ca or .uk
- Contrasting our results for bias with a more established concept such as **privacy**
- Expanding relevance criteria to include external links

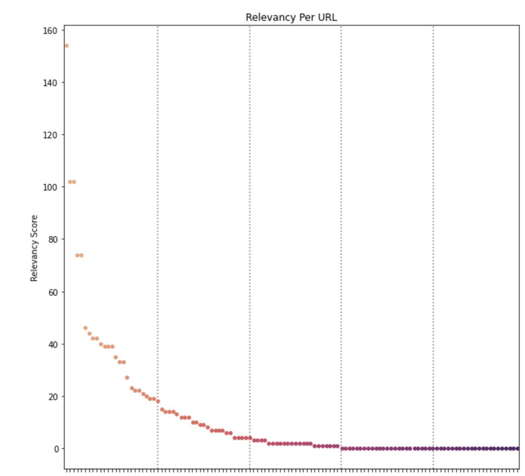

Figure 2. Relevance scores per URL. Analyzed scores to determine threshold of relevance (15), including the top quintile.

Title	Source	Goal	Suggested Practices
Blueprint for AI Bill of Rights: Algorithmic Discrimination Protections (2022)	White House Office of Science and Technology Policy (OSTP)	Prevent discrimination by algorithms and ensure systems are used and designed in an equitable way.	Proactive assessment of equity in design; Representative and robust data; Guarding against proxies; Ensuring accessibility during design, development, and deployment; Disparity assessment and mitigation; Ongoing monitoring and mitigation.
The Americans with Disabilities Act and the Use of Software, Algorithms, and Artificial Intelligence to Assess Job Applicants and Employees (2022)	U.S. Equal Employment Opportunity Commission	Protect individuals with disabilities against computer-based tools that assist in hiring workers, monitoring worker performance, determining pay or promotions, and establishing the terms and conditions of employment.	Provide situations in which employers handle people with disabilities in regards to computer based tools and how in certain scenarios those with disabilities can be affected by algorithms.
Algorithmic Justice and Online Platform Transparency Act (2021)	U.S. Senate	Prohibit the discriminatory use of personal information by online platforms in any algorithmic process, and require transparency in the use of algorithmic processes and content moderation, and for other purposes.	Prohibit algorithmic processes on online platforms that discriminate; Establish a safety and effectiveness standard for algorithms; Describe the types of algorithmic processes they employ and the information they collect; Maintain detailed records describing their algorithmic process; Publish annual public reports detailing their content moderation practices.
Towards a Standard for Identifying and Managing Bias in Artificial Intelligence (2022)	National Institute for Standards and Tech (NIST)	Root out bias in artificial intelligence and require addressing human and systemic biases.	Plan a series of public workshops at drafting a technical report for addressing AI bias and connecting the report with the AI Risk Management Framework.

Figure 3. Qualitative Analysis of Top 4 Most Relevant URLs. Out of the top 10 URLs, 4 URLs were relevant to our research focus. All URLs contained recommended practices and processes to mitigate algorithmic bias and discrimination.

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